Homework 2 - k Sample Inference

1 Question 1: Why do we Correct for Multiple Comparisons?

A student asked me what the point of correcting for multiple comparisons was. They made a two interesting points:

1. They are different outcomes, so, unlike the coin flip example (where the coin is the same) it doesn't makes sense to combine them.
2. If a manuscript presented three studies with t-tests, they don't correct for multiple comparisons across *those* tests.

When we conduct all pairwise comparisons after an omnibus test like ANOVA, we are testing a single family of hypotheses to answer one overarching question ("Which of these groups are different?"). Every additional test provides another opportunity to commit a Type I error from random noise. Similar to the coin, the probability of making at least one false error across J independent tests grows exponentially: $1-(1-\alpha)^J$. Note that our pairwise comparisons are not independent, so it's slightly different.

We don't correct for inferences across separate studies because they typically test distinct hypotheses using independent samples. Our attempt to address the multiple comparisons problem is specifically confined to the issue of increased Type I error within a single decision-making context. If we don't, we can find ourselves in a p-hacking-like situation, where an investigator might search through many tests to find one $p < 0.05$ (a potential false discovery), rather than making decisions to best position themselves for finding a true discovery.

Here, we will quickly demonstrate why we correct for the multiple comparisons problem when conducting several tests to answer the same question.

1.1 Part a

Complete the following steps in $R = 1000$ simulations.

1. Simulate data for $k = 5$ samples $n = 30$ drawn from a standard Gaussian distribution.
2. Use `pairwise.t.test(..., p.adjust.method="none")` to compute unadjusted pairwise t-tests and record if any comparison yields statistically discernible support for a non-zero mean.
3. Use `rstatix::tukey_hsd(...)` to compute the Tukey's HSD adjusted pairwise tests and record if any comparison yields statistically discernible support for a non-zero mean.

Solution

```
library(tidyverse)
library(rstatix)
set.seed(7272)

reject1 = rep(NA, 1000)
reject2 = rep(NA, 1000)

for (i in 1:1000) {
  sim <- data.frame(
    k1 = rnorm(30),
    k2 = rnorm(30),
    k3 = rnorm(30),
    k4 = rnorm(30),
    k5 = rnorm(30)) |>
```

```
    pivot_longer(cols = everything(), names_to = "Group", values_to = "Value")

  unadjusted = pairwise.t.test(sim$Value, sim$Group, p.adjust.method = "none")
  reject1[i] = any(unadjusted$p.value < 0.05, na.rm = TRUE)

  hsd = rstatix::tukey_hsd(sim, formula = Value ~ Group)
  reject2[i] = any(hsd$p.adj < 0.05, na.rm = TRUE)
}

mean(reject1)
```

```
[1] 0.286
```

```
mean(reject2)
```

```
[1] 0.054
```

In the end, describe the Type I error rates.

In 1000 simulations where all five groups were drawn from the same standard normal distribution, so the null hypothesis is true, the unadjusted pairwise t-tests produced at least one false positive in 28.6% of simulations. Each simulation involves 10 pairwise comparisons, each with its own 5% chance of a false positive, so the chances of at least one error accumulate. With Tukey's HSD, at least one comparison came out significant in 5% of the simulations, which is what we'd want. This shows why we correct for multiple comparisons: without it, we'd claim a difference in about 1 out of every 3 or 4 datasets when nothing is really going on.

1.2 Part b

Redo the simulation, changing to adjust according to the false discovery rate `p.adjust.method="BH"` and comment on the differences.

Solution

```
library(tidyverse)
library(rstatix)
set.seed(7272)

reject_none  = rep(NA, 1000)
reject_tukey = rep(NA, 1000)
reject_bh    = rep(NA, 1000)

for (i in 1:1000) {
  sim <- data.frame(
    k1 = rnorm(30, mean = 0, sd = 1),
    k2 = rnorm(30, mean = 0, sd = 1),
    k3 = rnorm(30, mean = 0, sd = 1),
    k4 = rnorm(30, mean = 0, sd = 1),
    k5 = rnorm(30, mean = 0, sd = 1)
  ) |>
    pivot_longer(cols = everything(), names_to = "Group", values_to = "Value")

  none = pairwise.t.test(sim$Value, sim$Group, p.adjust.method = "none")
  bh   = pairwise.t.test(sim$Value, sim$Group, p.adjust.method = "BH")
  hsd  = rstatix::tukey_hsd(sim, formula = Value ~ Group)

  reject_none[i]  = any(none$p.value < 0.05, na.rm = TRUE)
  reject_bh[i]    = any(bh$p.value   < 0.05, na.rm = TRUE)
  reject_tukey[i] = any(hsd$p.adj    < 0.05, na.rm = TRUE)
}
```

```
mean(reject_none)
```

```
[1] 0.286
```

```
mean(reject_tukey)
```

```
[1] 0.054
```

```
mean(reject_bh)
```

```
[1] 0.048
```

Redoing the simulation with bh still gives us a type I error of around 5%, except it is slightly smaller than the Tukey number. 4.8% compared to 5.4%.

2 Question 2: Power Analyses

Revisit the MFAP4 study from class.

2.1 Part a

Assume the researchers were targeting a moderate effect size of Cohen's $f = 0.25$, setting significance level $\alpha = 0.05$ and desired power $1 - \beta = 0.80$. Were they adequately powered?

Note: The actual effect size is $f = 0.55$, but researchers usually do not know that ahead of time and defaulted to moderate, instead of specifying a large effect.

What is the required sample size per group (n) and the overall total sample size (N)? Write the `R` code using the `pwr` package to calculate this sample size. This package assumes all samples are of the same size so $N = nk$, where k is the number of samples. Note that the best answers will have a corresponding graphical display.

Solution To detect a moderate effect size of f = 0.25 with 80% power we would need 40 observations per group with a N = 200 observations in total.

```
library(pwr)

power_result = pwr.anova.test(
  k = 5,
  f = 0.25,
  sig.level = 0.05,
  power = 0.80
)
power_result
```

```
     Balanced one-way analysis of variance power calculation

              k = 5
              n = 39.1534
              f = 0.25
      sig.level = 0.05
          power = 0.8

NOTE: n is number in each group
```

```
library(tidyverse)

# --- Power calculation under the N = 150 budget constraint ---
groups = 5
per_group = 30          # budget constraint: N = 150 / 5 groups
```

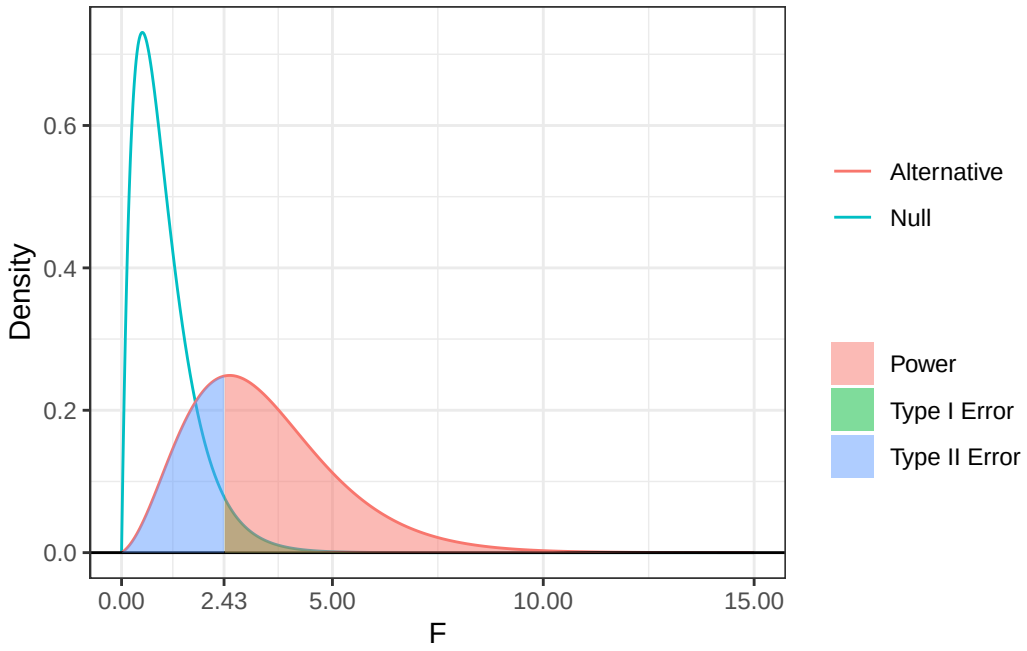
```r
eff_f = 0.25

num_df = groups - 1                      # numerator df = 4
den_df = groups * per_group - groups     # denominator df = 145
ncp_val = groups * per_group * eff_f^2   # non-centrality parameter = 9.375

crit_val = qf(0.95, num_df, den_df)

# --- Graphical display ---
plot_dat = tibble(fval = seq(0, 15, length.out = 1000)) |>
  mutate(dens_null = df(x = fval, df1 = num_df, df2 = den_df),
         dens_alt  = df(x = fval, df1 = num_df, df2 = den_df, ncp = ncp_val))

ggplot(plot_dat) +
  geom_line(aes(x = fval, y = dens_null, color = "Null")) +
  geom_line(aes(x = fval, y = dens_alt, color = "Alternative")) +
  geom_hline(yintercept = 0) +
  geom_ribbon(data = plot_dat |> filter(fval >= crit_val),
              aes(x = fval, ymin = 0, ymax = dens_null, fill = "Type I Error"), alpha = 0.5) +
  geom_ribbon(data = plot_dat |> filter(fval < crit_val),
              aes(x = fval, ymin = 0, ymax = dens_alt, fill = "Type II Error"), alpha = 0.5) +
  geom_ribbon(data = plot_dat |> filter(fval >= crit_val),
              aes(x = fval, ymin = 0, ymax = dens_alt, fill = "Power"), alpha = 0.5) +
  theme_bw() +
  scale_x_continuous("F", breaks = c(0, round(crit_val, 2), seq(5, 15, by = 5))) +
  labs(y = "Density", color = "", fill = "")
```



2.2 Part b

The project was funded by several agencies. Suppose that a funding agency suggested funding limiting the researchers to $N = 150$. Evaluate this suggestion in the context of the ANOVA. Would that be large enough to be adequately powered?

Compute the non-centrality parameter

$$\lambda = k \cdot n \cdot f^2$$

and use the non-central F distribution to compute the power under H_a for Cohen's $f = 0.25$. Calculate the statistical power achieved under this budget constraint. Note that the best answers will have a corresponding graphical display.

Solution This total sample size would not be large enough to be adequately powered. It would only have a power of .67 instead of .8 with n = 30 per group.

```
groups = 5
per_group = 30          # budget constraint: N = 150 / 5 groups
eff_f = 0.25

num_df = groups - 1                     # numerator df = 4
den_df = groups * per_group - groups    # denominator df = 145
ncp_val = groups * per_group * eff_f^2
ncp_val # non-centrality parameter = 9.375
```

```
[1] 9.375
```

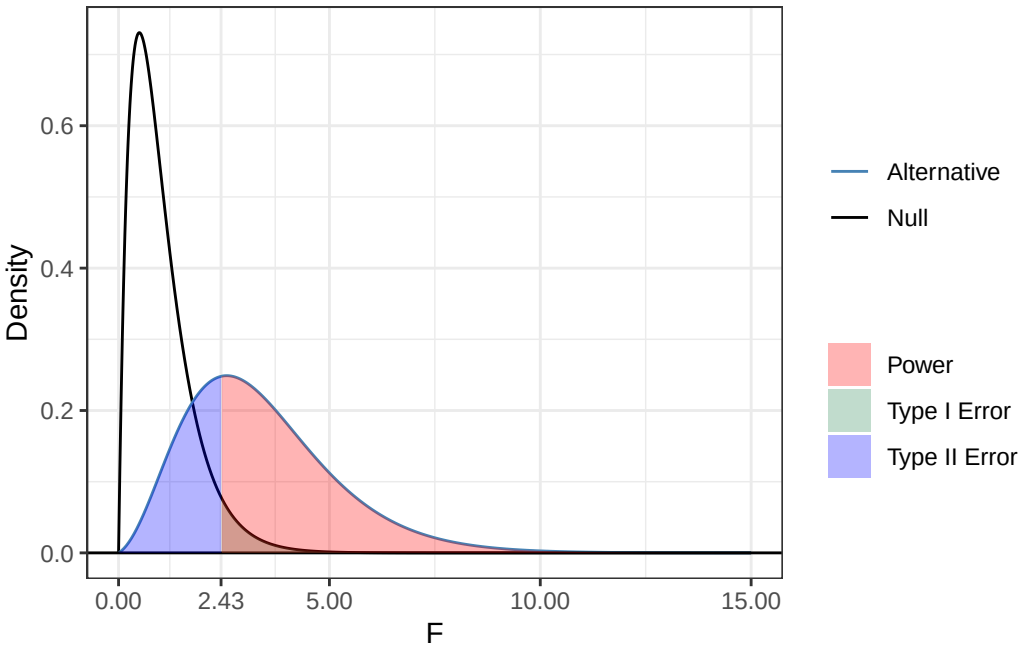
```
crit_val = qf(0.95, num_df, den_df)     # rejection threshold at alpha = 0.05

achieved_power = 1 - pf(crit_val, num_df, den_df, ncp = ncp_val)
achieved_power
```

```
[1] 0.6675964
```

```
plot_dat = tibble(fval = seq(0, 15, length.out = 1000)) |>
  mutate(dens_null = df(x = fval, df1 = num_df, df2 = den_df),
         dens_alt  = df(x = fval, df1 = num_df, df2 = den_df, ncp = ncp_val))

ggplot(plot_dat) +
  geom_line(aes(x = fval, y = dens_null, color = "Null")) +
  geom_line(aes(x = fval, y = dens_alt, color = "Alternative")) +
  geom_hline(yintercept = 0) +
  geom_ribbon(data = plot_dat |> filter(fval >= crit_val),
              aes(x = fval, ymin = 0, ymax = dens_null, fill = "Type I Error"), alpha = 0.3) +
  geom_ribbon(data = plot_dat |> filter(fval < crit_val),
              aes(x = fval, ymin = 0, ymax = dens_alt, fill = "Type II Error"), alpha = 0.3) +
  geom_ribbon(data = plot_dat |> filter(fval >= crit_val),
              aes(x = fval, ymin = 0, ymax = dens_alt, fill = "Power"), alpha = 0.3) +
  theme_bw() +
  scale_x_continuous("F", breaks = c(0, round(crit_val, 2), seq(5, 15, by = 5))) +
  scale_color_manual(values = c("Null" = "black", "Alternative" = "steelblue")) +
  scale_fill_manual(values = c("Type I Error" = "seagreen",
                               "Type II Error" = "blue",
                               "Power" = "red")) +
  labs(y = "Density", color = "", fill = "")
```



2.3 Part c

In the article the researchers state:

> *However, the statistical power was not sufficient to allow an accurate discrimination of no to moderate stages of hepatic fibrosis (F0–F2) from severe fibrosis and cirrhosis (F3 and F4).*

The European Association for the Study of the Liver (EASL) provides treatment recommendations based on these stages.

- Stages 0-1: Treatment is deferred with regularly assessment
- Stage 2: Treatment is justified for those in stage 2
- Stage 3-4: Treatment is a priority for those in stages

Suppose the researcher wanted to perform a follow-up test where they want to ensure adequate power not just for the overall ANOVA F-test, but specifically to detect a pairwise differences between stages 1 and 2. Explain why the sample size computed in Part a may not be enough and use `PMCMRplus::power.tukey.test()` to compute a sample size that attains adequate power.

Note: The researchers would have had to do this *a priori*. Because we aren't medical experts, you can use sample estimates using the original data.

$$
\begin{aligned}
n &= \text{answer from part a} \\
\mu_\Delta &= 0.25 \\
\sigma^2_{\text{pooled}} &= \frac{(176-1)0.85^2 + (135-1)0.84^2}{176+135-2} = 0.7152 \\
\sigma_{\text{within}} &= 0.77
\end{aligned}
$$

Solution The sample size calculated in part a would only result in a power of 70%. Using a while loop, we are able to calculate that the adequate number of observations for a power of 80% would be 315 per group.

```
library(PMCMRplus)
PMCMRplus::power.tukey.test(
  n = 40,
  groups = 5,
  delta = .25,
  within.var = .77,
```

```
  sig.level = 0.05
)
```

```
     Tukey-Test power calculation

              n = 40
         groups = 5
          delta = 0.25
     within.var = 0.77
          power = 0.07194129
    alternative = two-sided

NOTE: n is number in *each* group
```

```
n_current = 40
current_power = 0

while (current_power < 0.80) {
  n_current = n_current + 1
  result = PMCMRplus::power.tukey.test(
    n = n_current,
    groups = 5,
    delta = .25,
    within.var = .77,
    sig.level = 0.05
  )
  current_power = result$power
}

n_current
```

```
[1] 315
```

```
current_power
```

```
[1] 0.8006567
```

2.4 Part d

How does the answer to Part c change if they wanted to have adequate power for *all* pairwise comparisons? Use `Superpower::ANOVA_design()` to set up the ANOVA set up.

Note: Use design "5b" and the following.

$$\mu = \{2.92, 3.11, 3.36, 4.04, 4.35\}$$
$$\sigma^2 = \frac{(97-1)0.87^2 + (176-1)0.85^2 + (135-1)0.84^2 + (67-1)1.00^2 + (67-1)0.90^2}{97+176+135+67+67-5} = 0.80$$

Compute the power of the researcher's design (part a) using `Superpower::ANOVA_power()`. If the researchers were able to oversample by double, a common approach to addressing this issue, would their power be at least 80% for all pariwise comparisons?

Note: This code can take a minute. You may want to set `cache: true` in your `R` chunk yaml header. You will also see more power because the actual effect size was 0.55.

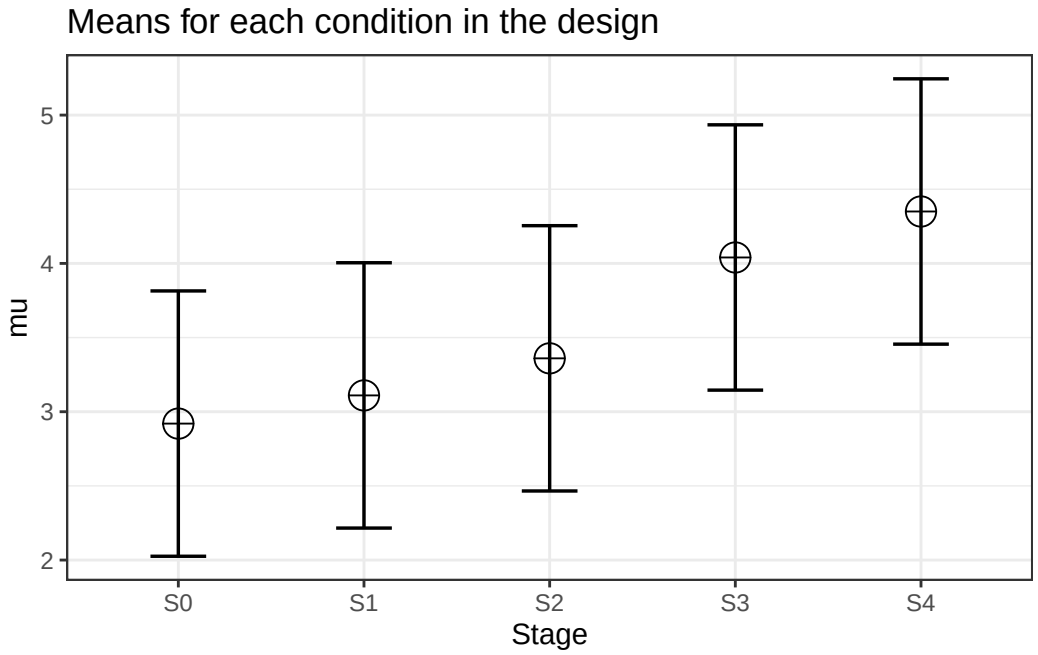
Solution If the researchers wanted adequate power for all pairwise comparisons, the required sample size would increase substantially. Instead of 40 observations per group as calculated in Part A, they would need 315 observations per group based on the Tukey power calculation in Part C. Doubling the sample size would not provide at least 80% power for all pairwise comparisons. Although the larger group would have more than the 315 observations per group

identified in Part C, the smaller group would only have 270 observations. Thus, the sample size would still be insufficient to achieve 80% power for all pairwise comparisons.

```r
library(Superpower)

group_means <- c(2.92, 3.11, 3.36, 4.04, 4.35)
pooled_sd <- sqrt(0.80)   # given variance is 0.80, ANOVA_design wants SD

design_result <- ANOVA_design(
  design = "5b",
  n = 40,               # part a's sample size per group
  mu = group_means,
  sd = pooled_sd,
  labelnames = c("Stage", "S0", "S1", "S2", "S3", "S4")
)
```




```r
power_result <- ANOVA_power(
  design_result,
  alpha_level = 0.05,
  nsims = 1000,
  seed = 7272
)
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Stage   100      0.2866

Power and Effect sizes for pairwise comparisons (t-tests)
                    power effect_size
p_Stage_S0_Stage_S1  16.3      0.2222
p_Stage_S0_Stage_S2  58.4      0.4928
p_Stage_S0_Stage_S3  99.9      1.2581
p_Stage_S0_Stage_S4 100.0      1.6124
p_Stage_S1_Stage_S2  22.0      0.2730
p_Stage_S1_Stage_S3  99.7      1.0409
p_Stage_S1_Stage_S4  99.9      1.3969
```

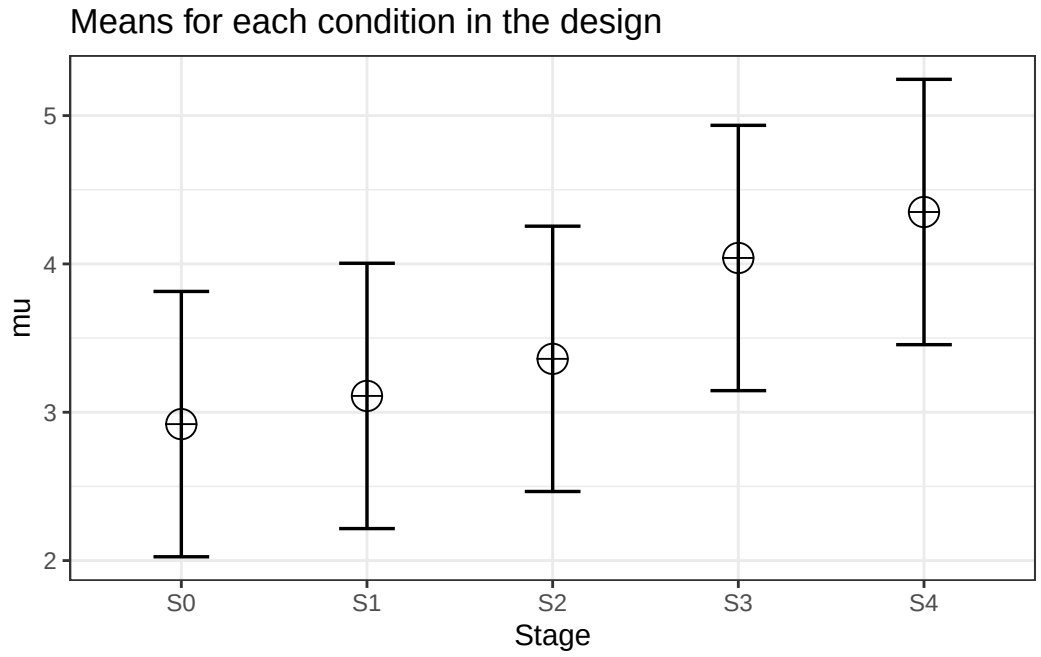
```
p_Stage_S2_Stage_S3  93.4      0.7670
p_Stage_S2_Stage_S4  99.9      1.1232
p_Stage_S3_Stage_S4  34.4      0.3584
```

```
power_result
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Stage   100      0.2866

Power and Effect sizes for pairwise comparisons (t-tests)
                    power effect_size
p_Stage_S0_Stage_S1  16.3      0.2222
p_Stage_S0_Stage_S2  58.4      0.4928
p_Stage_S0_Stage_S3  99.9      1.2581
p_Stage_S0_Stage_S4 100.0      1.6124
p_Stage_S1_Stage_S2  22.0      0.2730
p_Stage_S1_Stage_S3  99.7      1.0409
p_Stage_S1_Stage_S4  99.9      1.3969
p_Stage_S2_Stage_S3  93.4      0.7670
p_Stage_S2_Stage_S4  99.9      1.1232
p_Stage_S3_Stage_S4  34.4      0.3584
```

```
design_double <- ANOVA_design(
  design = "5b",
  n = 80,               # doubled from 40
  mu = group_means,
  sd = pooled_sd,
  labelnames = c("Stage", "S0", "S1", "S2", "S3", "S4")
)
```




```
power_double <- ANOVA_power(
  design_double,
  alpha_level = 0.05,
  nsims = 1000,
  seed = 7272
)
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Stage   100      0.2796

Power and Effect sizes for pairwise comparisons (t-tests)
                     power effect_size
p_Stage_S0_Stage_S1  25.9      0.2043
p_Stage_S0_Stage_S2  86.7      0.4908
p_Stage_S0_Stage_S3 100.0      1.2487
p_Stage_S0_Stage_S4 100.0      1.5976
p_Stage_S1_Stage_S2  42.5      0.2876
p_Stage_S1_Stage_S3 100.0      1.0477
p_Stage_S1_Stage_S4 100.0      1.3969
p_Stage_S2_Stage_S3  99.5      0.7580
p_Stage_S2_Stage_S4 100.0      1.1060
p_Stage_S3_Stage_S4  57.5      0.3486
```

```
1 power_double
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Stage   100      0.2796

Power and Effect sizes for pairwise comparisons (t-tests)
                     power effect_size
p_Stage_S0_Stage_S1  25.9      0.2043
p_Stage_S0_Stage_S2  86.7      0.4908
p_Stage_S0_Stage_S3 100.0      1.2487
p_Stage_S0_Stage_S4 100.0      1.5976
p_Stage_S1_Stage_S2  42.5      0.2876
p_Stage_S1_Stage_S3 100.0      1.0477
p_Stage_S1_Stage_S4 100.0      1.3969
p_Stage_S2_Stage_S3  99.5      0.7580
p_Stage_S2_Stage_S4 100.0      1.1060
p_Stage_S3_Stage_S4  57.5      0.3486
```

3 Question 3: An Application

Yang et al. (2023) investigate whether low-dose nicotine can counteract age-related metabolic decline. The proposed mechanism is through NAD^+, which is a chemical compound essential for metabolic energy production. The idea is that low-dose nicotine could help improve and/or maintain the recycling of NAD^+ by activating the salvage enzyme NAMPT via SIRT1. This could, in turn, restore NAD^+ homeostasis, which is the biological balance maintained within a cell between the production, recycling, and degradation of nicotinamide adenine dinucleotide (NAD^+).

Note: It is low-dose nicotine. Cigarette smoking delivers massive doses of nicotine along with other harmful chemicals. This is not an endorsement for smoking.

Male mice across distinct life stages – 2, 6, and 16 months – were categorized to evaluate natural, age-dependent metabolic and enzymatic shifts in two conditions.

- Treatment: low-dose nicotine (administered in drinking water)
- Control: no-dose placebo (administered in drinking water)

The researchers then aimed to compare treatment against control in the following samples:

- 2 months old, control
- 6 months old, control
- 16 months old, control
- 16 months old, treatment

Using PET imaging, the researchers measured age-induced inflammatory glucose hypermetabolism (SUV_{max}). They want to show that this quantity increases with age and those in the nicotine group saw levels comparable to younger mice.

Table 1: SUV_{max} measurements in the heart across age and treatment groups

Replicate	2M	6M	16M	16M, Treatment
1	1.48969	4.36000	7.30182	4.69304
2	1.30168	4.28807	7.12416	2.51326
3	1.30620	3.64184	8.28764	2.38019

3.1 Part a

State the null (H_0) and alternative (H_a) hypotheses for the omnibus test.

Solution

H_0: $\mu_{2M} = \mu_{6M} = \mu_{16M}$ (control) $= \mu_{16M}$

H_a: at least one differs from the others.

3.2 Part b

Perform some exploratory data analysis to provide empirical information about the researchers question.

```
library(tidyverse)
library(knitr)

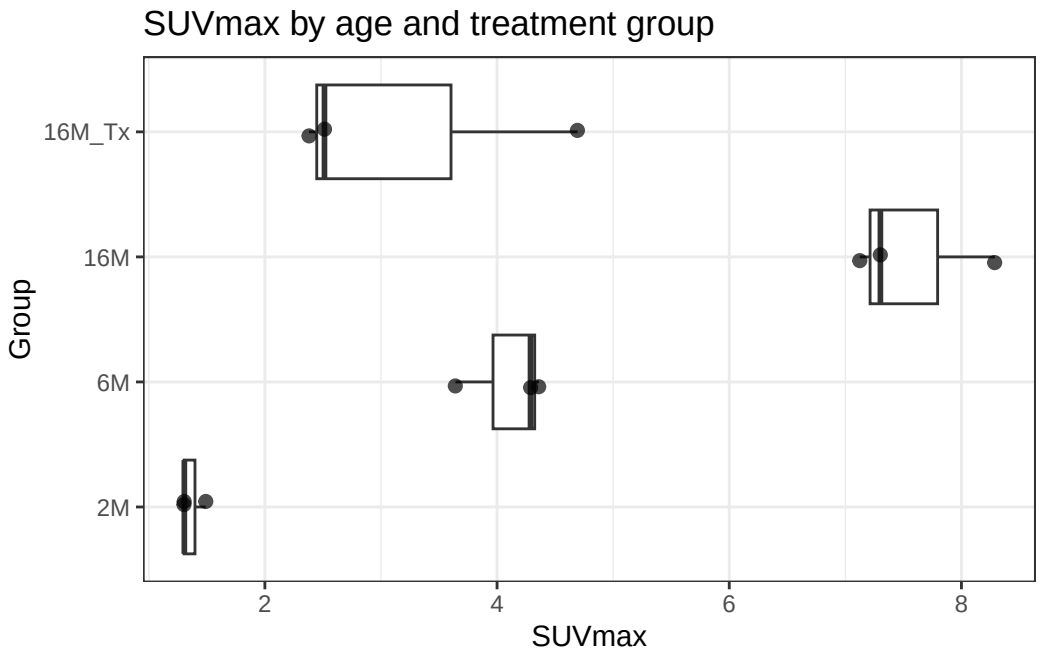
suv_data = tibble(
  group = rep(c("2M", "6M", "16M", "16M_Tx"), each = 3),
  suvmax = c(1.48969, 1.30168, 1.30620,
             4.36000, 4.28807, 3.64184,
             7.30182, 7.12416, 8.28764,
             4.69304, 2.51326, 2.38019)
) |>
  mutate(group = factor(group, levels = c("2M", "6M", "16M", "16M_Tx")))

suv_data |>
  group_by(group) |>
  summarize(
    n = n(),
    mean = mean(suvmax),
    median = median(suvmax),
    sd = sd(suvmax),
    min = min(suvmax),
    max = max(suvmax)
  ) |>
  kable(digits = 2, caption = "SUVmax by Age and Treatment Group")
```

Table 2: SUVmax by Age and Treatment Group

group	n	mean	median	sd	min	max
2M	3	1.37	1.31	0.11	1.30	1.49
6M	3	4.10	4.29	0.40	3.64	4.36
16M	3	7.57	7.30	0.63	7.12	8.29
16M_Tx	3	3.20	2.51	1.30	2.38	4.69

```r
ggplot(suv_data, aes(x = suvmax, y = group)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(height = 0.05, size = 2, alpha = 0.7) +
  theme_bw() +
  labs(x = "SUVmax", y = "Group",
       title = "SUVmax by age and treatment group")
```



3.3 Part c

Perform and report the results of the omnibus test. Make sure to evaluate the conditions for using the method you choose.

Solution We cannot assume normality as there is only 3 observations per group, although the number of observations per group is the same. Because the variances are not equal, I used a welch's ANOVA test as opposed to a standard ANOVA. We can see statistically discernable support for the alternative as the p value is $0.000685 < 0.05$, meaning at least one of the groups is different.

```r
library(effectsize)

welch_result = oneway.test(suvmax ~ group, data = suv_data, var.equal = FALSE)
welch_result
```

```
    One-way analysis of means (not assuming equal variances)

data:  suvmax and group
F = 98.464, num df = 3.0000, denom df = 3.5496, p-value = 0.0006851
```

```r
omega_squared(welch_result)
```

```
# Effect Size for ANOVA

Omega2 |       95% CI
---------------------
0.97   | [0.83, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

3.4 Part d

Conduct post-hoc pairwise comparisons to determine if age-induced inflammatory glucose hypermetabolism increase with age and whether low-dose nicotine reduces age-induced inflammatory glucose hypermetabolism.

Solution

```
library(rstatix)
library(knitr)

games_howell_result = games_howell_test(suv_data, suvmax ~ group)

games_howell_result |>
  kable(digits = 4, caption = "Games-Howell Pairwise Comparisons for SUVmax")
```

Table 3: Games-Howell Pairwise Comparisons for SUVmax

.y.	group1	group2	estimate	conf.low	conf.high	p.adj	p.adj.signif
suvmax	2M	6M	2.7308	1.3080	4.1535	0.0119	*
suvmax	2M	16M	6.2054	3.8141	8.5966	0.0072	**
suvmax	2M	16M_Tx	1.8296	-3.3049	6.9641	0.3083	ns
suvmax	6M	16M	3.4746	1.5618	5.3873	0.0080	**
suvmax	6M	16M_Tx	-0.9011	-5.4738	3.6715	0.6978	ns
suvmax	16M	16M_Tx	-4.3757	-8.5077	-0.2437	0.0431	*

3.5 Part e

Write a short paragraph directly addressing the researcher's hypothesis using your work in (a)-(d).

Overall, this analysis supports the researchers' hypothesis. Since the group variances weren't equal, I used Welch's ANOVA for the omnibus test, and it showed a significant difference in SUVmax across the four groups ($F = 98.5$, $p = 0.00068$, $^2 = 0.97$). This means age and/or nicotine treatment are affecting glucose hypermetabolism in the heart. Looking at the Games-Howell pairwise comparisons, SUVmax goes up significantly with age in the control mice: 2M vs 6M ($p = 0.012$), 6M vs 16M ($p = 0.008$), and 2M vs 16M ($p = 0.007$). So age-induced inflammatory glucose hypermetabolism really does increase over the mouse lifespan, like the researchers expected. The more interesting part is the treatment group. SUVmax in the 16-month treated mice was significantly lower than in the 16-month untreated mice ($p = 0.043$), and wasn't significantly different from the 2-month ($p = 0.308$) or 6-month ($p = 0.698$) groups.

References

Yang, Liang, Junfeng Shen, Chunhua Liu, et al. 2023. "Nicotine Rebalances NAD+ Homeostasis and Improves Aging-Related Symptoms in Male Mice by Enhancing NAMPT Activity." *Nature Communications* 14 (1): 900.